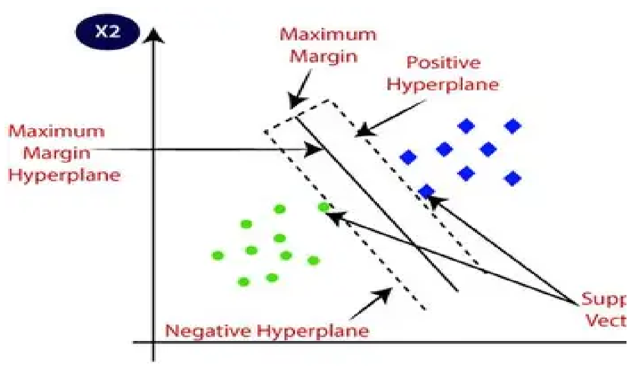
Overview of Classification Algorithms in Machine Learning

1. Support Vector Machine (SVM):

Description:

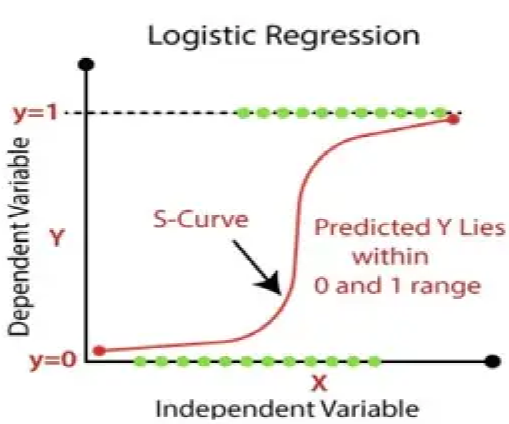
Support Vector Machine is a supervised learning algorithm used mainly for classification tasks. It works by finding the best boundary (called a hyperplane) that separates data into different classes. The goal is to maximize the margin between the closest points of different classes. It can also handle non-linear data using kernel tricks. SVM is effective in high-dimensional spaces



2. Logistic Regression :

Description:

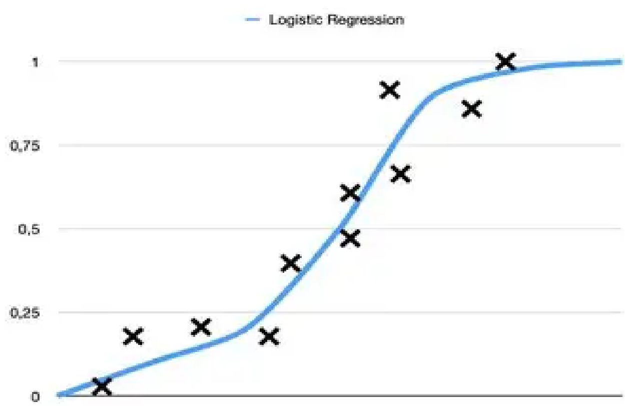
Logistic Regression is used for classification problems, especially binary classification. It predicts the probability that an input belongs to a certain class using a sigmoid function. The output is always between 0 and 1, making it easy to interpret as probability. Despite its name, it is used for classification, not regression. It is simple, fast, and works well for linearly separable data.



3. Naive Bayes:

Description:

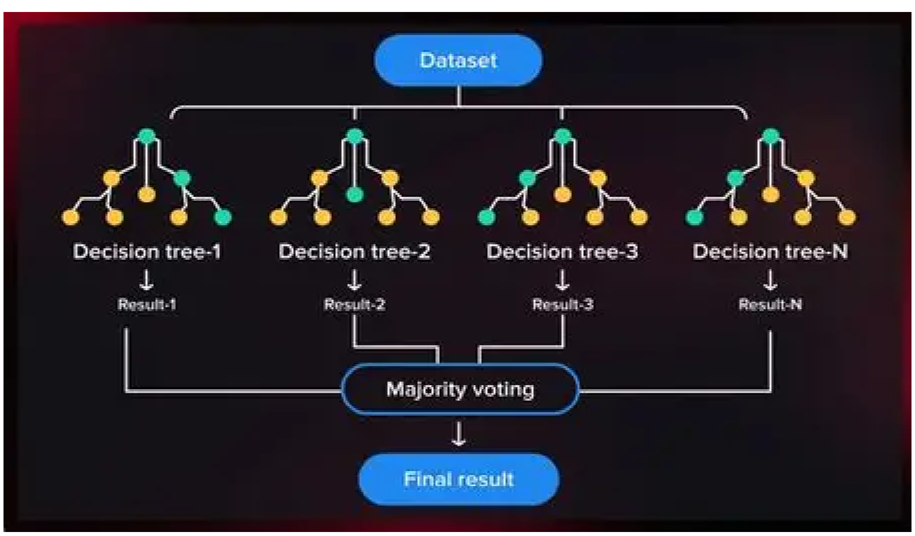
Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem. It assumes that all features are independent of each other (which is a "naive" assumption). It calculates the probability of each class and chooses the one with the highest probability. It is very fast and works well with large datasets. It is commonly used in text classification like spam detection



4. Decision Tree Classifier:

Description:

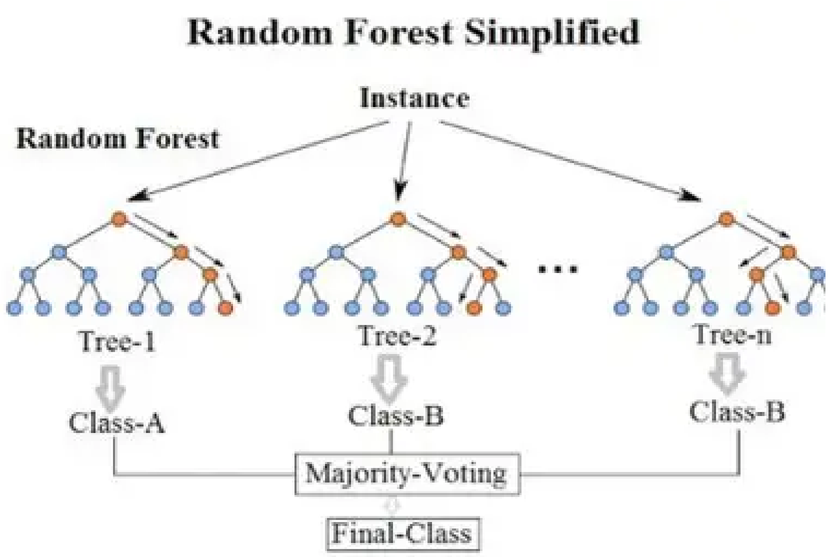
A Decision Tree is a model that makes decisions based on a tree-like structure. It splits the data into branches based on feature values, leading to a final decision at the leaves. Each internal node represents a condition, and each branch represents an outcome. It is easy to understand and visualize. However, it can overfit if the tree becomes too deep.



5. RandomForest Classifier:

Description:

Random Forest is an ensemble learning method that combines multiple decision trees. Each tree is trained on a random subset of the data and features. The final prediction is made by majority voting among all trees. It reduces overfitting and improves accuracy compared to a single decision tree. It is powerful and widely used in real-world problems



Final Conclusion:

All these algorithms are used to sort data into categories, but they work in different ways. Some use chances or probabilities (Naive Bayes, Logistic Regression), while others draw lines to separate data (SVM) or use tree-like steps (Decision Tree, Random Forest). Choosing the best algorithm depends on the type of data, how much data you have, and how complex the problem is. Methods like Random Forest, which combine many models, usually give better results.

I learned :

how different machine learning algorithms put data into categories in different ways. I understood ideas like probability-based models, decision boundaries, and tree-based methods. I also learned that choosing the right model depends on the problem. Overall, I now have a basic understanding of five important classification algorithms used in machine learning.